Marvin Gülhan

Education

University of Bonn — Bonn, Germany
Master of Science in Computer Science · Track: Intelligent Systems & Algorithms — *Oct. 2025 – present*

University of Bonn — Bonn, Germany
Bachelor of Science in Computer Science (Grade: 1.2) — *Oct. 2022 – Sep. 2025*

Rhein-Sieg-Gymnasium — Sankt Augustin, Germany
Abitur (Grade: 1.0) — *Aug. 2014 – June 2022*

German International School of Silicon Valley — Mountain View (CA), USA
Semester Abroad — *Aug. 2019 – Jan. 2020*

Work Experience & Internships

Research Assistant — Wachtberg, Germany
Fraunhofer Institute FKIE — *Nov. 2023 – present*
- Algorithm engineering for online graph problems
- Research on RL algorithms for dynamic planning problems

Formula Student Team Member — Sankt Augustin, Germany
Bonn-Rhein-Sieg University of Applied Sciences (HBRS) — *Sep. 2023 – Jun. 2025*
- Member of the Formula Student team at HBRS
- CFD simulation and aerodynamic design of the rear wing for the 2024 race car

Student Internship — Cologne, Germany
German Aerospace Center (DLR) — *Jan. 2020*
- Insights into aerospace engineering and research methods
- Use of clustering methods for anomaly detection in fuel combustion of hybrid engines

Publications & Projects

Reinforcement Learning Amplifies Emergent Misalignment from Harmless Rewards — 2026
- Co-author of a paper submitted to the 2026 Empirical Methods in Natural Language Processing (EMNLP), Budapest, Hungary
- Shows that reinforcement learning (GRPO, preceded by a brief SFT warmup) can broadly misalign a language model not only when rewarding overtly harmful behavior but also when the reward signal looks harmless. Suggests misspecified rewards are a realistic path to misalignment.

Multi-Agent Reinforcement Learning for Complex and Dynamic Graph-Based Planning Problems — 2026
- Co-author of a peer-reviewed paper accepted at the 2026 International Conference on Military Communications and Information Systems (ICMCIS), Bath, UK
- Demonstrated that Graph Reinforcement Learning policies can effectively solve complex, dynamic planning problems such as the Airlift Planning Problem (in-press)

Airlift Challenge: A Competition for Optimizing Cargo Delivery — 2025
- Co-author of a research paper on optimizing cargo delivery for Airlift Operations
- Developed a routing algorithm coordinating heterogeneous aircraft across subgraphs to manage bottlenecks and disruptions in dynamic conditions, achieving 2nd place out of 40 teams

Honors

Deutschlandstipendium — Bonn, Germany
Federal Ministry of Education and Research — *Oct. 2024 & Oct. 2025*
- Merit-based scholarship for outstanding academic performance

Core Skills

Preferred Languages: Python, C++
Technical Skills: Algorithm Engineering · Machine Learning · Operations Research · Combinatorial Optimization · Reinforcement Learning · Online Algorithms · Research Engineering · LLM Fine-Tuning & Evaluation
Languages: German (native), English (C1), French (B1), Spanish (B1)